DATE	28-02-2026
TEAM ID	LTVIP2026TMID*****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	2 MARKS

2.1 PROBLEM STATEMENT

Accurate classification of blood cells is a critical step in diagnosing various hematological disorders such as anemia, leukemia, infections, and immune-related diseases. Traditionally, blood cell analysis is performed manually by medical experts using microscopic examination. This process is time-consuming, labor-intensive, and highly dependent on the expertise and experience of laboratory professionals. Human errors, fatigue, and variability in interpretation can lead to inconsistent or inaccurate results.

With the increasing number of patients and growing demand for faster diagnostic services, manual analysis alone is no longer sufficient to ensure efficiency and precision. Additionally, developing deep learning models from scratch requires large datasets, high computational resources, and extensive training time, which may not always be feasible.

In many healthcare settings, especially in resource-limited areas, access to skilled hematologists may be limited. Delays in diagnosis can affect timely treatment and patient outcomes. There is also a need for standardized and consistent analysis across laboratories to avoid diagnostic variations.

Therefore, there is a need for an automated, accurate, and time-efficient system that can classify blood cells reliably. HematoVision addresses this problem by utilizing transfer learning techniques to improve classification accuracy while reducing training time and computational cost, ultimately supporting healthcare professionals in making faster and more consistent diagnostic decisions.